

ProTox 3.0：用于化学品毒性预测的网络服务器

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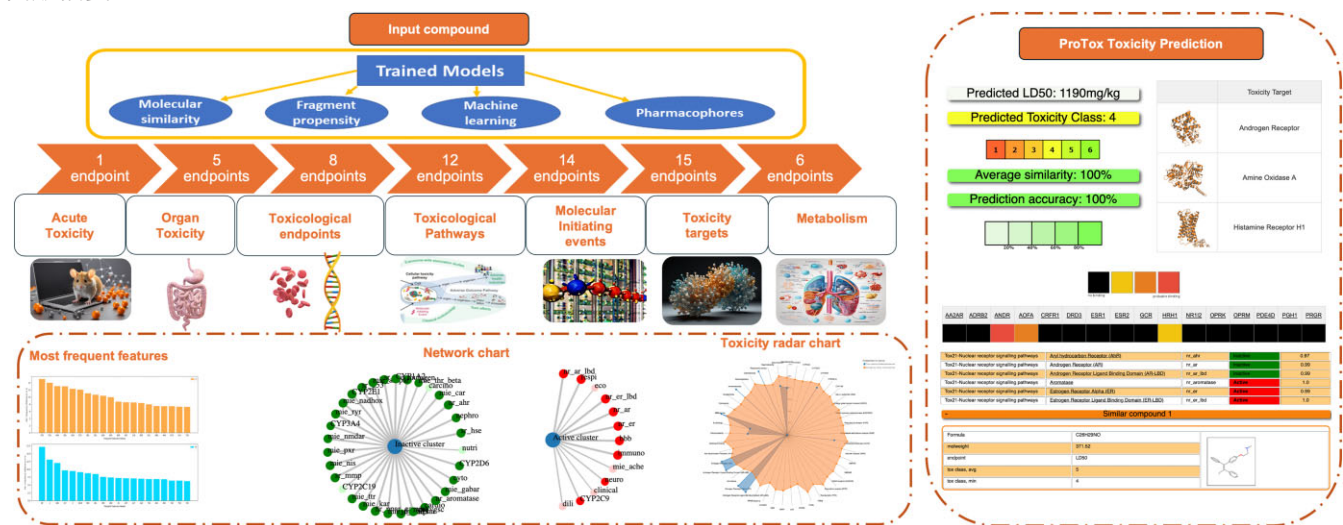
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摘要

与存在于药物、食品、环境和消费品中的化学物质的相互作用，是我们日常生活中不可或缺的一部分。然而，根据剂量和持续时间不同，这些相互作用也可能导致不良影响。随着计算方法的增加，计算机模拟（in silico）方法能够为法规需求、风险评估以及制药行业评估化学品安全性提供显著的益处。在此，我们介绍ProTox 3.0，它结合了分子相似性和机器学习模型，用于预测61个毒性终点，例如急性毒性、器官毒性、临床毒性、分子起始事件（MOE）、不良结果（Tox21）通路、其他若干毒理学终点及毒性靶点。所有ProTox 3.0模型都在独立的外部数据集上进行了验证，并显示出强劲的性能。ProTox设想自身为一个完整、免费可用的计算工具。

图形摘要



介绍

毒性评估是药物和化学品设计的主要步骤之一。因此，对于计算预测模型以评估药物和化学品的潜在毒性作用的需求很高。监管决策机构如欧洲药品管理局（EMA）和美国食品药品监督管理局（FDA），以及环境健康保护机构如美国环境保护署

美国环境保护局（EPA）和欧洲环境署（EEA）支持绿色化学范式，并了解绿色毒理学的重要性（1）。绿色毒理学原则的重要步骤包括——更安全的设计，以及早期测试以生产更安全的化学品并使测试过程可持续（2）。毒性测试是降低对人类和环境风险的前提条件。绿色毒理学方法的一个重要方面是减少

ProTox 3.0: a webserver for the prediction of toxicity of chemicals

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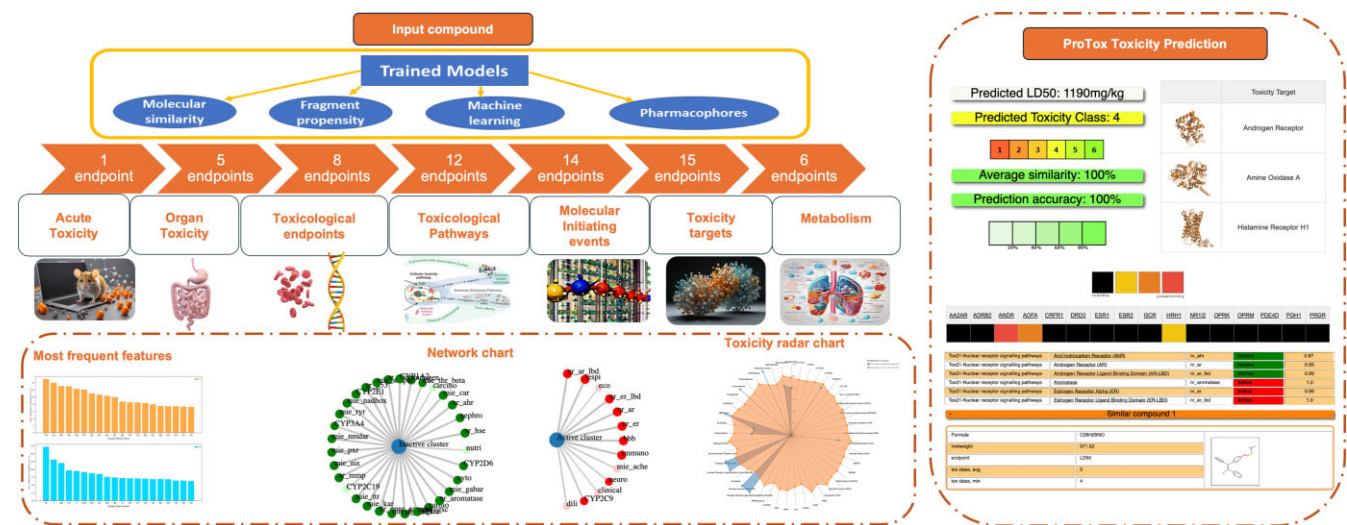
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Abstract

Interaction with chemicals, present in drugs, food, environments, and consumer goods, is an integral part of our everyday life. However, depending on the amount and duration, such interactions can also result in adverse effects. With the increase in computational methods, the *in silico* methods can offer significant benefits to both regulatory needs and requirements for risk assessments and the pharmaceutical industry to assess the safety profile of a chemical. Here, we present ProTox 3.0, which incorporates molecular similarity and machine-learning models for the prediction of 61 toxicity endpoints such as acute toxicity, organ toxicity, clinical toxicity, molecular-initiating events (MOE), adverse outcomes (Tox21) pathways, several other toxicological endpoints and toxicity off-targets. All the ProTox 3.0 models are validated on independent external sets and have shown strong performance. ProTox envisages itself as a complete, freely available computational platform for *in silico* toxicity prediction for toxicologists, regulatory agencies, computational chemists, and medicinal chemists. The ProTox 3.0 webserver is free and open to all users, and there is no login requirement and can be accessed via <https://tox.charite.de>. The web server takes a 2D chemical structure as input and reports the toxicological profile of the compound for each endpoint with a confidence score and overall toxicity radar plot and network plot.

Graphical abstract



Introduction

Toxicity evaluation is one of the main steps in drug and chemical design. Hence, there is a high demand for computational predictive models to evaluate the potential toxic effects of drugs and chemicals. The regulatory decision-making agencies such as the European Medicines Agency (EMA), and U.S. Food and Drug Administration (FDA), Environmental health protection agencies like the U.S. Environmental Protection

Agency (EPA) and the European Environment Agency (EEA) to support the green chemistry paradigm and understand the importance of green toxicology (1). Important steps of green toxicology principles include- safer designs, and early testing to produce safer chemicals and make the testing process sustainable (2). Toxicity testing is a prerequisite for reducing the risk to humans and the environment. One of the important aspects of the green toxicology approach is reducing the

在毒性测试过程中动物实验和化学品使用的数量 (3)。

据估计, 在新药开发过程中, 通常会考虑平均约 10,000 至 25,000 种化学物质。总体而言, 临床前毒性 (动物) 和不良事件 (人体毒性) 占药物候选物淘汰案例的 ~1/3 (4)。

绿色毒理学实践促进和鼓励采用减少动物实验使用、减少测试过程中使用和处理的化学品数量的新技术和创新策略; 并增加在化学品的合成、使用和监管中对毒性的考虑。绿色化学专注于设计能够最小化或消除有害物质、减少废物产生、节约能源和资源的化学产品和工艺。通过将绿色化学原理纳入化学合成和制造, 绿色毒理学有助于开发更安全且对环境更友好的化学品。通常, 仍需评估许多正在考虑的物质, 并且用于测试的物质和资源数量有限。这些限制要求使用计算方法和高通量体外方法。智能测试策略, 如计算机模拟、高通量

因此, 计算机模拟毒性正在发展成为一种预测各种化学物质对人类、动物、植物和生态系统潜在危害的综合平台。计算机模拟毒性模型的目标是补充现有的体外毒性方法, 以预测化学物质的毒性效应, 从而优化用于毒性评估的时间和财务资源。计算机模拟毒性模型融合了毒理学、生物统计学、系统生物学、计算机科学及许多其他相关学科的知识。本质上, 计算机模拟毒性建模代表了一种复杂且全面的化学安全评估方法, 有望彻底改变我们评估和减轻化学物质暴露风险的方式。

一个好的预测模型的前提条件取决于敏感性和特异性之间的性能平衡。敏感性是预测给定数据集中所有有毒 真阳性的能力, 而特异性是预测的能力

正确地预测非毒性 真阴性 (6)。ProTox 3.0模型在考虑各自模型敏感性和特异性的情况下显示出平衡的性能比。ProTox 3.0网站服务器中报告的所有61个模型的性能都比商业软件如Discovery Studio的TOPKAT (计算机辅助技术毒性预测; Accelrys, Inc., 美国) (<http://accelrys.com/>) 以及美国环境保护局开发的免费工具毒性估算软件工具 (T.E.S.T.) (<https://www.epa.gov/chemical-research/toxicity-estimation-software-tool-test>) 表现相对更好。

此外, 网络服务器的出现, 其中一些向更广泛的社区提供部分访问权限, 带来了显著的益处。admetSAR平台拥有一系列基于QSAR方法的预测模型 (47个模型)。ProTox——网络服务器的以前版本 (8,9) 已被广泛引用, 并且在实际验证中被发现有 用 (10,11)。ProTox平台已被作为若干大学的讲座模块使用。使用ProTox服务器获得的结果和分析也会定期在如欧洲化学品管理局 (ECHA) 等监管平台上报告, 作为动物实验替代方法的补充报告 (12)。

ProTox 3.0 网络服务器相较于现有的计算模型提供了多项优势。ProTox 网络服务器包括化学、分子靶点知识、代谢以及不良反应途径 (AOPs)。正如我们在早期更新 (ProTox-II) 中提到的 (8), ProTox 网络服务器的一大新颖之处在于, 该预测方案被分类为不同级别的毒性, 例如口服毒性、器官毒性 (肝毒性、神经毒性、呼吸毒性、心脏毒性和肾毒性)、毒理终点 (如诱变性、致癌性、细胞毒性、免疫毒性、血脑屏障渗透性、生态毒性、临床毒性和营养毒性)、12 条毒理途径 (AOPs)、15 个毒性靶点、14 个分子起始事件 (MIEs) 靶点以及 6 个用于代谢的分子靶点, 从而提供关于此类毒性反应潜在分子机制的见解。最新版本 ProTox 3.0 包含了分子

ProTox 3.0 平台

输入信息

ProTox 3.0 网络服务器的用户友好界面信息自足, 使用时几乎不需要帮助或完全不需要帮助。它的设计以用户自我直观的视角为出发点。用户需要指定输入化合物的名称 (PubChem 名称) 或规范 SMILES (简化分子输入线性表示系统) 字符串, 以在服务器上运行预测。此外, 用户还可以使用由 (<https://www.chemdoodle.com/>) 提供的绘图框绘制所需的分子结构。这甚至支持在实验室合成之前对假想化合物进行预测。此外, 用户可以选择单个模型/ 用户特定模型或所有模型; 如果用户未指定, 网络服务器默认仅计算急性毒性和毒性靶点。

number of animal tests and chemical usage during toxicity testing (3).

It has been estimated that an average of between 10 000 and 25 000 individual chemicals are considered in the course of the development of a new drug. Together, pre-clinical toxicity (animal) and adverse events (human toxicity) account for ~1/3 of the cases of attrition of drug candidates (4).

Green Toxicology practices promote and encourage the use of new and innovative techniques and strategies that reduce the use of animals for testing, the amounts of chemicals used and disposed of during tests; and increase the consideration of toxicity in the synthesis, use and regulation of chemicals. Green chemistry focuses on the design of chemical products and processes that minimize or eliminate hazardous substances, reduce waste generation, and conserve energy and resources. By incorporating green chemistry principles into chemical synthesis and manufacturing, Green Toxicology contributes to the development of safer and more environmentally benign chemicals. Often, many more substances still under consideration must be assessed, and that quantities of substances and resources for testing are limited. These limitations call for the use of computational and higher-throughput *in vitro* approaches. Intelligent testing strategies, such as *in silico* modeling, high-throughput screening and structure–activity relationship analysis, enable efficient and cost-effective identification of potential toxicity concerns. By leveraging these predictive tools, researchers can prioritize safer alternatives and design products with reduced environmental and health impacts (3). Green Toxicology seamlessly integrates with the principles of Green Chemistry, forming a cohesive approach to sustainable chemical design and usage (5). By acknowledging and addressing the potential adverse outcomes and toxicity associated with chemical development, utilization, and disposal, Green Toxicology reinforces the overarching goals of Green Chemistry. Toxicological tools, including *in silico* modeling, omics technologies, and *in vitro* methods, play a pivotal role in enhancing our understanding of toxicity mechanisms and identifying structural–activity relationships (SAR) and associated effects. With the help of these tools, researchers can often identify and eliminate potentially harmful chemical candidates at an early stage, a concept often referred to as ‘failing early and failing cheaply’ (3). Through a shared emphasis on understanding and mitigating toxicity, Green Toxicology and Green Chemistry foster a holistic approach to sustainable chemical innovation (2,3).

Thus, *in silico*, toxicity is evolving as an integral platform for the prediction of the potential hazards posed by various chemical substances to humans, animals, plants and ecosystems (6). The aim of *in silico* toxicity models is to complement the existing *in vitro* toxicity methods to predict the toxicity effects of chemicals, thereby streamlining both time and financial resources allocated for toxicity assessments. *In silico* toxicity model incorporates knowledge from various fields such as toxicology, biostatistics, systems biology, computer science and many other relevant disciplines (5). In essence, *in silico* toxicity modeling represents a sophisticated and comprehensive approach to evaluating chemical safety, holding the promise to revolutionize how we assess and mitigate the risks inherent in chemical exposure.

A good prediction model’s prerequisites depend on the performance balance between sensitivity and specificity. Sensitivity is the ability to predict all the toxic/true positives in the given dataset, whereas specificity is the ability to pre-

dict the non-toxic/true negatives correctly (6). ProTox 3.0 models have shown a balanced performance ratio considering the sensitivity and specificity of the respective models. All the 61 models reported in the ProTox 3.0 web server showed relatively better performance than commercial software like Discovery Studio’s TOPKAT (Toxicity Prediction by Komputer Assisted Technology; Accelrys, Inc., USA) (<http://accelrys.com/>) as well as freely accessible tool like Toxicity Estimation Software Tools (T.E.S.T.) developed by the U.S. Environmental Protection Agency (<https://www.epa.gov/chemical-research/toxicity-estimation-software-tool-test>).

Additionally, the emergence of web servers, some of which offer partial access to the wider community, has brought significant benefits. The admetSAR platform, which houses a range of prediction models rooted (47 models) in QSAR methods (7). ProTox- previous versions of the web servers (8,9) have been highly cited and have been found useful in real-world validation(10,11). ProTox platform has been used as lecture module in several universities. The results and analysis obtained using the ProTox server are also regularly reported on the regulatory platform like European Chemicals Agency (ECHA) as additional reports on alternative methods to animal trials (12).

The ProTox 3.0 web server provides several advantages over existing computational models. ProTox webserver includes chemical, molecular target knowledge, metabolism, and the Adverse outcomes pathways (AOPs). As mentioned in our earlier update (ProTox-II) (8), a novelty of the ProTox webserver, in general, is that the prediction scheme is classified into different levels of toxicity such as oral toxicity, organ toxicity (hepatotoxicity, neurotoxicity, respiratory toxicity, cardiotoxicity, and nephrotoxicity), toxicological endpoints (such as mutagenicity, carcinogenicity, cytotoxicity, immunotoxicity, BBB-permeability, ecotoxicity, clinical toxicity and nutritional toxicity), 12 toxicological pathways (AOPs), 15 toxicity targets, 14 targets for molecular initiating events (MIEs), and six molecular targets for metabolism—thereby providing insights into the possible molecular mechanism behind such toxic response. The latest version, ProTox 3.0, incorporates molecular similarity, pharmacophore-based, fragment propensities, and machine learning models for the prediction of various toxicity endpoints. ProTox 3.0 consisting of 61 models, is a freely available computational toxicity prediction webserver enabling the prediction of the most toxicological endpoints to date.

ProTox 3.0 platform

Input information

The user-friendly interface of the ProTox 3.0 web server is self-informative and can be used with less or no help required. It is designed with a self-intuitive user perspective in mind. The user needs to specify either the name (PubChem name) or canonical SMILES (*Simplified Molecular-Input Line Entry System*) string of the input compound to run a prediction on the server. Furthermore, the user can also use the drawing box provided by the (<https://www.chemdoodle.com/>), to draw user -desired molecular structure. This supports the prediction of even a hypothetical compound before synthesizing it in the lab. Additionally, the user can select a single model/user-specific model or ALL models; if not specified by the user, the web server computes only acute toxicity and toxicity targets by default.

输出信息

预测输入化合物的结果页面包括急性毒性等级的信息（预测的半数致死量 (LD50) 单位为 mg/ /kg 体重、毒性等级及预测准确性），包括数据集中存在的三种最相似化合物的三种结构，以及输入化合物相对于其他数据库分子的理化性质分布。随后是一个表格，包含器官毒性、毒理学终点、毒理学途径、分子起始事件和代谢的预测结果及其各自的预测分类和置信评分。结果页面还包括预测的毒性靶点信息，包括靶点名称、输入化合物相对于药效团和已知配体的平均拟合度和相似度。用户可以指定特定感兴趣的模型或所有模型进行预测。预测结果也以毒性雷达图的形式显示活性类别预测。

材料与方法

更新后的 ProTox 3.0 平台分为七个不同的分类步骤：(i) 急性毒性（口服毒性模型，包含六种不同的毒性等级）；(ii) 器官毒性（五个模型）；(iii) 毒理学终点（8 个模型）；(iv) 毒理学途径（12 个模型）；(v) 分子起始事件（14 个模型）；(vi) 代谢（六个模型）；(vii) 毒性靶点（15 个模型）。在此，我们简要描述了 ProTox 3.0 服务器上可用的每个新模型，并提及了已发表的模型（8）。有关详细信息，包括参考文献、性能评分以及训练集中最常见特征（对于有毒/类和无毒/类分子）的频率分布，可在 ProTox 3.0 服务器的模型信息中获取。本研究中使用的数据集数量的完整描述也可在服务器的“模型信息”下查看。ProTox 3.0 更新目前共报道了 61 个模型。

新开发的28个模型用于ProTox 3.0网络服务器，基于随机森林（RF）机器学习和深度神经网络算法以及八种不同的数据采样方法（13）。使用基于RF的分类器的优势在于它可以避免过拟合，并且比基于深度学习的模型表现更好。使用了两种不同的分子指纹：MACCS分子指纹（MACCS结构键；Accelrys：美国加利福尼亚州圣地亚哥，2011。<http://accelrys.com/>）和Morgan环形指纹（<http://www.rdkit.org/>）（14,15）。然而，修改后的MACCS指纹在大多数验证集中表现略好。

关于各自模型、数据集和采样方法的更多信息，可通过网络服务器的“模型信息”和“常见问题”部分获取。

预测模型是基于 Python 编程语言的。像 scikit-learn 这样的机器学习包

（<http://scikit-learn.org>）和化学信息学包 RDKit（<http://www.rdkit.org/>）被用于模型实现。所有数据使用 KNIME (29) 进行了标准化。在 ProTox 3.0 网页服务器的常见问题部分的“使用 API”描述下提供了一个模板脚本（示例 API 脚本 https://comptox.charite.de/prottox3/prottox3_api.py）。ProTox-3.0 数据存储 in 关系型 MySQL 数据库中。MyChem 包用于处理数据库中的化学信息。对其大多数功能而言，MyChem 依赖于 Open Babel 工具箱。网站后台使用 PHP 构建；通过 Apache HTTP Server 实现网页访问。Redis 是一个用于排队和评估 API 请求的高效键-值存储。

所有新模型均使用10折交叉验证进行验证。数据按照8种不同的采样方法进行了划分，确保活性和非活性比例在10个集合中保持不变，其中9个用于训练模型，第10个用于验证模型。通过sklearn库的RandomizedSearchCV函数找到了分类器的最佳参数。此外，还使用了一个外部数据集（模型未知）进行模型的外部验证。所有模型的评估指标包括准确率、受试者工作特征曲线（ROC）的曲线下面积（AUC）、敏感性、特异性以及F1值来衡量二分类模型的质量。F1值范围在0（较差）到1（完美）之间（16,17）。总共构建了224个模型用于28个终点（每个终点8个模型），并选择各终点表现最好的模型作为最终模型，并将其保存为模块。

急性毒性

口服毒性

急性毒性模型是通过分析已知有毒化合物之间的化学相似性并识别特定的毒性片段构建的。这一方法在我们先前的出版物中有所阐述 (9)。为了填充急性毒性模型，我们从内部数据库 SuperToxic 的更新版本中提取了相关数据 (18)。

毒性靶点

毒性靶点预测涉及对来自著名诺华体外安全面板（19）的15个不同靶点的分析。这些特定的蛋白靶点与不良药物反应密切相关，相关方法在我们早期的出版物中有详细讨论（8）。

器官毒性

肝毒性

药物性肝损伤模型基于我们先前发表的文献中报道的方法（8,13）

神经毒性

药物引起的神经毒性可能导致大脑正常功能的严重并发症，从而引发各种神经系统症状或疾病（12）。中枢神经系统，包括大脑和脊髓，特别容易受到神经毒性物质的影响。计算和机器学习模型有可能提供更高效、更具成本效益的方法来识别神经毒性

Output information

The result page of the predicted input compound includes information on the acute toxicity class (predicted median lethal dose (LD₅₀) in mg/kg weight, toxicity class, and prediction accuracy), including the three structures of the three most similar compounds present in the dataset, and distribution of physicochemical properties of the input compound compared to other database molecules. This is followed by a table containing the prediction for the organ toxicity, toxicological endpoints, toxicological pathways, molecular initiating events, and metabolism with their respective predicted class and confidence score. The results page also includes the predicted toxicity targets with information on the target’s name, average fit and similarity of the input compound concerning pharmacophore and known ligands. Users can specify a specific model of interest or all models for the prediction. The prediction results are also shown as a toxicity radar plot for active class prediction. The active and inactive classes’ results are also displayed as a network plot for an input compound (Figure 1). The new server allows users to download the results in several file formats- csv, pdf, image files. More detailed information with an example compound output is available in the ‘FAQs’ section of the ProTox 3.0 webserver.

Materials and methods

The updated ProTox 3.0 platform is divided into seven different classification steps: (i) acute toxicity (oral toxicity model with six different toxicity classes); (ii) organ toxicity (five models); (iii) toxicological endpoints (8 models); (iv) toxicological pathways (12 models), (v) molecular initiating events (14 models), (vi) metabolism (six models) and (vii) toxicity targets (15 models). Here, we briefly describe each new model available on the ProTox 3.0 server and mention the already published models (8). Detailed information with references, performance scores and frequency distribution of the most common features present in the training set (both for toxic/positive class and non-toxic/negative class) molecule is available under *model info* on the ProTox 3.0 webserver. A complete description of the number of data sets used in this study is provided under ‘Model info’ of the server. There are currently a total of 61 models reported in the ProTox 3.0 update. A total of 28 new models have been added to the recent version, and the previous (33) models have also been updated based on the new server software specification.

The newly developed 28 models to the ProTox 3.0 webserver are based on Random Forest (RF) machine learning and Deep neural network algorithm and eight different data sampling methods (13). The advantage of using an RF-based classifier is that it avoids overfitting and performs better than the deep learning-based models. Two different molecular fingerprints are used: MACCS molecular fingerprints (MACCS Structural keys; Accelrys: San Diego, CA, 2011. <http://accelrys.com/>) and Morgan circular fingerprints (<http://www.rdkit.org/>) (14,15). However, modified MACCS fingerprints performed slightly better on most of the validation sets.

More information on the respective models, datasets, and sampling methods is available via the ‘Model info’ and ‘FAQs’ sections of the web server.

The prediction models are based on Python programming language. Machine learning packages like scikit-learn

(<http://scikit-learn.org>) and cheminformatics package RDKit (<http://www.rdkit.org/>) are used for the model implementation. All data are standardised using KNIME (29). A template script (sample API script https://comptox.charite.de/prottox3/prottox3_api.py) has been provided under the description ‘using the API’ on the FAQ section of the ProTox 3.0 webserver. ProTox-3.0 data is stored in a relational MySQL database. The MyChem package is used to handle the chemical information within the database. For most of its functions, MyChem relies on the Open Babel toolbox. The website backend is built using PHP; web access is enabled via the Apache HTTP Server. Redis is an agile key/value store for queueing and assessing API requests.

All the new models are validated using 10-fold cross-validation. The data was divided accordingly by 8 different sampling methods, ensuring the active and inactive ratios were constant into 10 sets, 9 of which were used to train the model and 10th to validate the model. The best parameters for the classifier were found through the Randomized-SearchCV function of the sklearn library. Additionally, an external set (unknown to the model) was used for external validation of the models. All the models are assessed on the following performance—accuracy, the area under the curve (AUC) of a receiver operating characteristic (ROC), sensitivity, specificity and F1 measures the quality of binary classification models. The F1 ranges between 0 (less significant) to 1 (perfect) (16,17). A total of 224 models for 28 endpoints were constructed (8 models for single endpoint), and the best-performing model for respective endpoints was selected as the final model, saved with the module pickle and implemented on the ProTox 3.0 server.

Acute toxicity

Oral toxicity

The acute toxicity model is constructed by analysing chemical similarities among compounds with known toxic effects and identifying specific toxic fragments. This methodology was elaborated on in our prior publication (9). To populate the acute toxicity model, we have extracted relevant data from the updated version of our in-house database, SuperToxic (18).

Toxicity targets

Toxicity target prediction lies in the analysis of 15 distinct targets sourced from the renowned Novartis *in vitro* safety panels (19). These specific protein targets are intricately linked to adverse drug reactions, the methods are extensively discussed in our earlier publication (8).

Organ toxicity

Hepatotoxicity

Drug-induced liver injury model is based on the method reported in our previous publication (8,13)

Neurotoxicity

Drug-induced neurotoxicity can result in severe complications in the normal functioning of the brain, leading to various neurological symptoms or disorders (12). The central nervous system, including the brain and spinal cord, is particularly vulnerable to the effects of neurotoxic substances. Computational and machine learning models can potentially provide more efficient and cost-effective ways to identify neurotox-

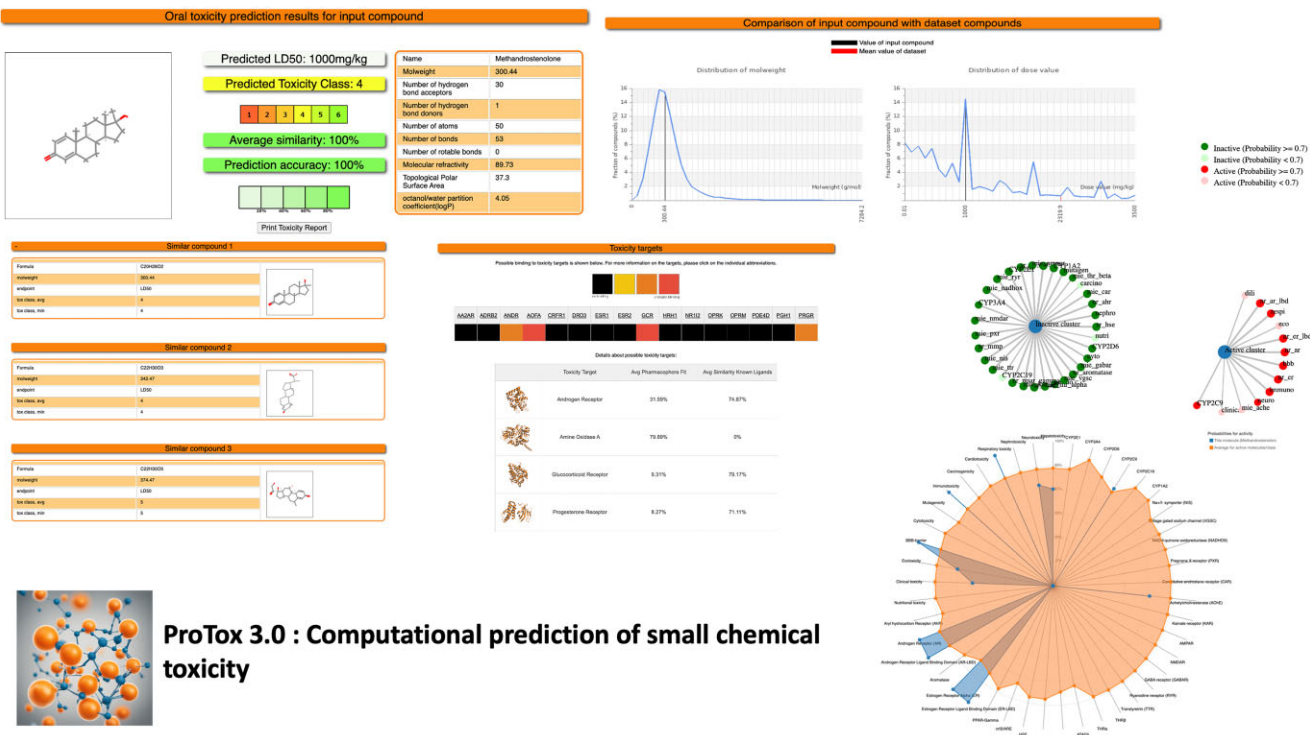


图 1. 应用案例：使用示例化合物“甲基睾丸素”（已撤回药物）。

与临床药物相关的神经毒性。已知神经毒性和非神经毒性药物的数据对于训练精确模型至关重要，这些数据是从文献中整理的 (12, 13, 20)。SMOTE-VDM——数据采样方法 (13) 被用于构建模型。ProTox 3.0 预测模型在交叉验证集上的平衡准确率为86.00%，在外部验证集上的平衡准确率为80.00%。交叉验证和外部验证的AUC-ROC得分分别为0.87和0.91（表1和表2）。

肾毒性

药物诱导的肾毒性可能发生在肾脏暴露于外源性或内源性毒物时。暴露于肾毒性药物可能会破坏肾脏维持体内稳态的功能。肾毒性可能通过多种机制发生，例如结晶性肾病、血栓性微血管病、肾小管细胞毒性和炎症 (21)。计算和机器学习模型可能提供更高效且更具成本效益的方法来识别与临床药物相关的神经毒性。已知肾毒性和非肾毒性药物的数据对于训练准确模型至关重要，这些数据已从文献中整理而来 (21)。kMedoids2 数据采样方法 (13) 被用于构建模型。ProTox 3.0 预测模型在交叉验证集上的平衡准确率为 83.00%，在外部验证集上的平衡准确率为 78.00%。交叉验证和外部验证的 AUC-ROC 分别为 0.86 和 0.89（表 1 和表 2）。

心脏毒性

药物诱导的心脏毒性可能由抑制人以太-去-去相关基因（hERG）通道引起——该钾离子通道在调节心脏动作电位中起着关键作用 (22, 23)。hERG通道对于心脏细胞的复极化是必不可少的，其抑制

某些小分子可能导致心电图（ECG）上QT间期延长。ProTox-3.0心脏毒性模型预测基于小分子的hERG阻断剂 (22)。关于心脏毒性和非心脏毒性化合物的数据——一个包含5252种具有hERG抑制值（IC50和Ki）的化合物的数据集——来自ChEMBL数据库 (24)。SMOTETC数据采样方法 (13) 用于构建模型。ProTox 3.0预测模型在交叉验证集上的平衡准确率为87.00%，在外部验证集上的平衡准确率为89.00%。交叉验证和外部验证的AUC-ROC评分分别为0.86和0.95（表1和表2）。

呼吸毒性

药物诱导的呼吸毒性是指某些药物或药物暴露引起的对呼吸系统的不良影响。接触某些化学物质和药物可能导致副作用，对呼吸系统产生重大影响，从轻度呼吸不适到严重呼吸窘迫的症状。已使用公开数据库整理已知呼吸毒性和非呼吸毒性药物的数据 (25)。使用随机过采样方法 (randOS) (13) 数据采样方法来构建模型。ProTox 3.0 预测模型在交叉验证集上的准确率为 91.90%，在外部验证集上的准确率为 88.90 %。交叉验证和外部验证的 AUC-ROC 分别为 0.98 和 0.93 (表 1 和表 2)。

毒理学终点

ProTox 3.0 网络服务器目前在毒理学终点类别下托管八个模型。

四种模型，即致癌性、致突变性、细胞毒性和免疫毒性模型，已经在我们之前发表的研究中报告 (8)。更多相关信息关于

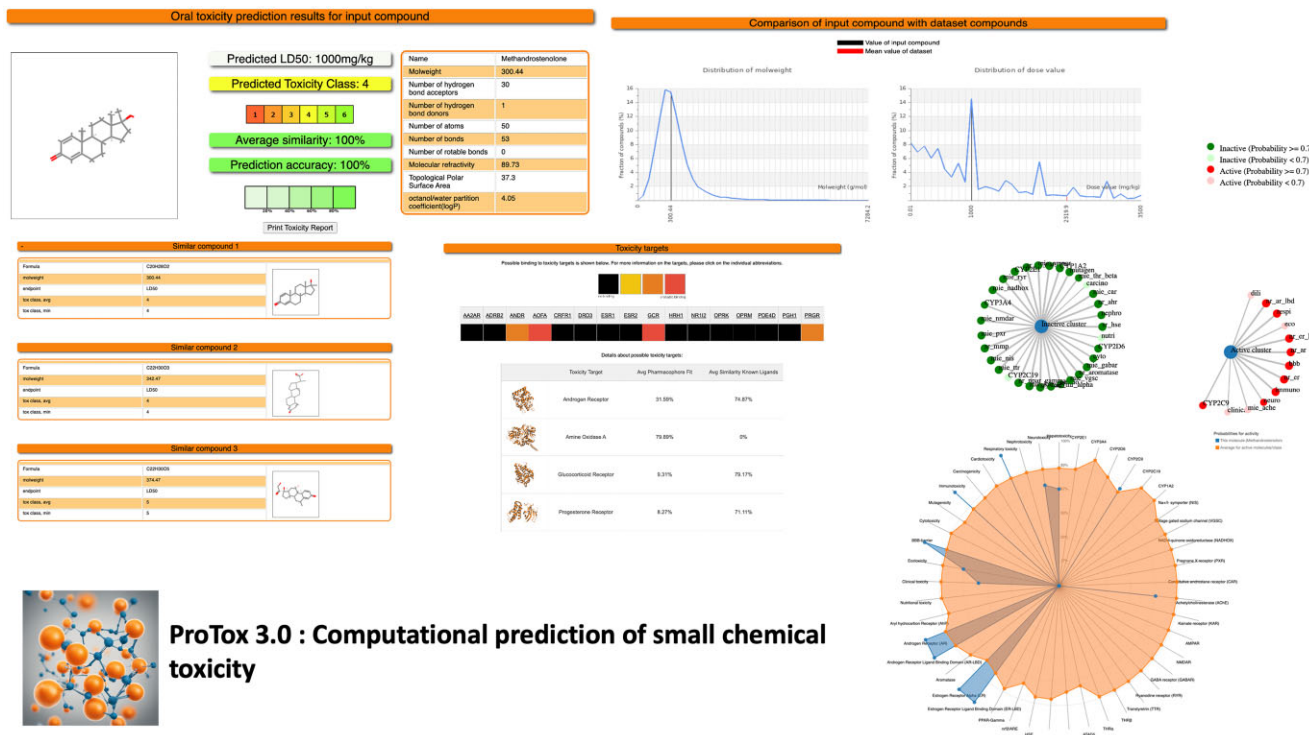


Figure 1. Application case: Using an example compound ‘methandrostenolone’ (withdrawn drug).

icity associated with clinical drugs. Data on known neurotoxic and non-neurotoxic drugs is crucial for training accurate models and was curated from the literature (12,13,20). SMOTEVDM—the data sampling method (13) was used to construct the model. The ProTox 3.0 prediction model has a balanced accuracy of 86.00% on the cross-validation set and 80.00% on an external validation set. The AUC-ROC scores of cross-validations and external validation are 0.87 and 0.91, respectively (Tables 1 and 2).

Nephrotoxicity

Drug-induced nephrotoxicity can occur when the kidney is exposed to exogenous or endogenous toxicants. Exposure to nephrotoxic drugs can destabilize the kidney’s function of maintaining homeostasis of the body. Nephrotoxicity can occur due to various mechanisms, such as crystal nephropathy, thrombotic microangiopathy, tubular cell toxicity, and inflammation (21). Computational and machine learning models can potentially provide more efficient and cost-effective ways to identify neurotoxicity associated with clinical drugs. Data on known nephrotoxic and non-nephrotoxic drugs is crucial for training accurate models and was curated from the literature (21). kMedoids2—the data sampling method (13) was used to construct the model. The ProTox 3.0 prediction model has a balanced accuracy of 83.00% on the cross-validation set and 78.00% on an external validation set. The AUC-ROC scores of cross-validations and external validation are 0.86 and 0.89, respectively (Tables 1 and 2).

Cardiotoxicity

Drug-induced cardiotoxicity can be caused by inhibiting the human ether-à-go-go-related gene (hERG) channel- a potassium ion channel that plays a crucial role in regulating the cardiac action potential (22,23). The hERG channel is essential for the repolarization of the cardiac cells, and its inhibition

by certain small molecules can lead to the prolongation of the QT interval on an electrocardiogram (ECG). The ProTox-3.0 cardiotoxicity model predicts small molecules-based hERG blockers (22). Data on cardiotoxic and non-cardiotoxic chemical compounds a dataset of 5252 compounds with hERG inhibition values (IC50 and Ki) was obtained from the ChEMBL database (24). SMOTETC- the data sampling method (13) was used to construct the model. The ProTox 3.0 prediction model has a balanced accuracy of 87.00% on the cross-validation set and 89.00% on an external validation set. The AUC-ROC scores of cross-validations and external validation are 0.86 and 0.95, respectively (Tables 1 and 2).

Respiratory toxicity

Drug-induced respiratory toxicity refers to adverse effects on the respiratory system caused by exposure to certain drugs or medications. Exposure to certain chemicals and drugs can result in side effects that can have a major impact on the respiratory system, leading to symptoms ranging from mild respiratory discomfort to severe respiratory distress. Data on known respiratory toxic and respiratory non-toxic drugs were curated using publicly available databases (25). The random oversampling method (randOS) (13) the data sampling method was used to construct the model. The ProTox 3.0 prediction model has an accuracy of 91.90% on the cross-validation set and 88.90% on an external validation set. The AUC-ROC scores of cross-validations and external validation are 0.98 and 0.93, respectively (Tables 1 and 2).

Toxicological endpoints

ProTox 3.0 webserver currently hosts eight models under the toxicological endpoints category.

Four models, namely carcinogenicity, mutagenicity, cytotoxicity, and immunotoxicity models are already reported in our previous published studies (8). More information on the

表1. 在Protox 3.0平台中新加入模型的交叉验证集上的性能评估, 包括准确率、敏感性、特异性和AUC-ROC

模型 (交叉验证)	准确率 (%)	AUC-ROC	敏感性 (%)	特异性 (%)
器官毒性				
神经毒性	86.00	0.87	82.80	86.90
肾毒性	83.00	0.86	78.70	86.80
心脏毒性	87.00	0.86	86.00	95.10
呼吸毒性	91.80	0.98	90.00	87.50
毒理学终点				
血脑屏障 (BBB)	98.00	0.96	92.00	97.00
生态毒性	90.00	0.95	92.00	93.00
临床毒性	86.00	0.84	82.40	79.50
营养毒性	83.80	0.84	80.11	90.00
分子启动事件 (MIE)				
THR α	94.20	0.97	95.70	92.80
THR β	95.00	0.97	90.90	95.40
TTR	86.00	0.93	86.60	85.30
RYR	97.00	0.98	94.80	96.80
加巴	89.20	0.95	86.00	92.30
NMDAR	98.00	0.99	98.00	97.50
AMPA受体	97.00	0.95	98.70	96.00
KAR	96.00	0.94	90.00	98.00
AChE	92.60	0.95	93.50	89.90
CAR	94.00	0.98	91.90	98.00
PXR	81.30	0.90	78.90	85.50
纳多克斯	96.90	0.95	96.70	97.10
VGSC	91.40	0.95	91.50	93.00
NIS	99.00	0.98	98.00	99.00
新陈代谢				
CYP1A2	95.00	0.97	99.00	97.00
CYP2C19	97.00	0.97	99.00	94.00
CYP2C9	97.00	0.98	97.00	96.00
CYP2D6	84.00	0.92	88.00	86.00
CYP3A4	92.00	0.96	93.00	92.00
CYP2E1	90.00	0.95	98.00	97.00

各自的模型、数据点、特征重要性和性能矩阵可以通过服务器上的模型信息获取。

在当前的更新中, ProTox 3.0 预测工作流程中新增了用于预测血脑屏障 (BBB) 渗透性、生态毒性、临床毒性和营养毒性的开发模型。

血脑屏障 (BBB)

一些毒素由于其大小、亲脂性或特定的运输机制, 能够绕过或破坏血脑屏障 (BBB) 并进入大脑 (26)。这些有毒物质可能包括环境污染物、药物以及某些化学物质。虽然临床实验是预测分子血脑屏障通透性最准确的方法, 但另一方面, 它们耗时且劳动强度大。关于血脑屏障通透性的正负类别数据集对于训练准确模型至关重要, 并已从文献中整理 (27)。采用SMOTETC——数据采样方法 (13) 来构建模型。ProTox 3.0预测模型在交叉验证集上的平衡准确率为98.00%, 在外部验证集上的平衡准确率为97.00%。交叉验证和外部验证的AUC-ROC评分分别为0.96和0.91 (表1和表2)。

生态毒性

在 ProTox 3.0 平台上开发的生态毒性模型可以很好地预测扩展化学空间领域对鱼类、水蚤和藻类的急性毒性。已知有毒和无毒化合物的数据对于训练精确模型至关重要, 这些数据是从文献中整理的 (28, 29)。SMO TEVDM——数据采样

采用方法 (13) 构建了该模型。ProTox 3.0 预测模型在交叉验证集上的平衡准确率为90.00%, 在外部验证集上的平衡准确率为81.20%。交叉验证和外部验证的AUC-ROC分别为0.95和0.85 (表1和表2)。

临床毒性

本研究中报告的临床毒性模型终点被定义为药物类分子是否因毒性而在临床试验中失败。数据集来自ClinTox数据集 (30)。已知的临床毒性药物和临床非毒性药物的数据 分子基于毒性报告的存在被定义为撤回 失败。kMedoids2——数据抽样方法 (13) 被用于构建模型。ProTox 3.0预测模型在交叉验证集上的平衡准确率为86.00%, 在外部验证集上的平衡准确率为79.00%。交叉验证和外部验证的AUC-ROC评分分别为0.84和0.87 (表1和表2)。

营养毒性

从公开可获得的数据库中获取了与食品相关化学物质的高质量毒理学数据 (31, 32)。这些数据被整理为食品中存在的有毒和无毒化学物质, 最终的数据集用于训练和验证。使用kMedoids1 数据采样方法 (13) 来构建模型。ProTox 3.0 预测模型在交叉验证集上的平衡准确率为83.80%, 在外部验证集上的平衡准确率为84.70%。交叉验证和外部验证的AUC-ROC分别为0.84和0.85 (表1和表2)。

Table 1. Performance evaluation on the cross-validation set for the newly included models in the ProTox 3.0 platform in terms of accuracy, sensitivity, specificity and AUC-ROC

Models (cross-validation)	Accuracy (%)	AUC-ROC	Sensitivity (%)	Specificity (%)
Organ toxicity				
Neurotoxicity	86.00	0.87	82.80	86.90
Nephrotoxicity	83.00	0.86	78.70	86.80
Cardiotoxicity	87.00	0.86	86.00	95.10
Respiratory toxicity	91.80	0.98	90.00	87.50
Toxicological endpoints				
Blood–brain barrier (BBB)	98.00	0.96	92.00	97.00
Ecotoxicity	90.00	0.95	92.00	93.00
Clinical toxicity	86.00	0.84	82.40	79.50
Nutritional toxicity	83.80	0.84	80.11	90.00
Molecular initiating events (MIE)				
THR α	94.20	0.97	95.70	92.80
THR β	95.00	0.97	90.90	95.40
TTR	86.00	0.93	86.60	85.30
RYR	97.00	0.98	94.80	96.80
GABAR	89.20	0.95	86.00	92.30
NMDAR	98.00	0.99	98.00	97.50
AMPAR	97.00	0.95	98.70	96.00
KAR	96.00	0.94	90.00	98.00
AChE	92.60	0.95	93.50	89.90
CAR	94.00	0.98	91.90	98.00
PXR	81.30	0.90	78.90	85.50
NADHOX	96.90	0.95	96.70	97.10
VGSC	91.40	0.95	91.50	93.00
NIS	99.00	0.98	98.00	99.00
Metabolism				
CYP1A2	95.00	0.97	99.00	97.00
CYP2C19	97.00	0.97	99.00	94.00
CYP2C9	97.00	0.98	97.00	96.00
CYP2D6	84.00	0.92	88.00	86.00
CYP3A4	92.00	0.96	93.00	92.00
CYP2E1	90.00	0.95	98.00	97.00

respective models, data points, feature importance, and performance matrix is available on the server via Model Info.

In the current update, newly developed models for predicting the blood-brain barrier (BBB) permeability, ecotoxicity, clinical toxicity, and nutritional toxicity have been added to the ProTox 3.0 prediction workflow.

Blood–brain barrier (BBB)

Some toxins, by size, lipophilicity, or specific transport mechanisms, can bypass or compromise the BBB and enter the brain (26). These toxicants may include environmental pollutants, drugs, and certain chemicals. Though clinical experiments are the most accurate methods to predict the BBB permeability of a molecule, they are, on the other hand, time-consuming and labour-intensive. Data sets on the positive and negative classes of the BBB permeability are crucial for training accurate models and were curated from the literature (27). SMOTETC—the data sampling method (13) was used to construct the model. The ProTox 3.0 prediction model has a balanced accuracy of 98.00% on the cross-validation set and 97.00% on an external validation set. The AUC-ROC scores of cross-validations and external validation are 0.96 and 0.91, respectively (Tables 1 and 2).

Ecotoxicity

The ecotoxicity model developed on the ProTox 3.0 platform can predict the acute toxicity of the extended chemical space domain on fish, daphnids, and algae with good great prediction accuracy. Data on known toxic and non-toxic compounds is crucial for training accurate models and was curated from the literature (28,29). SMOTEVDM—the data sampling

method (13) was used to construct the model. The ProTox 3.0 prediction model has a balanced accuracy of 90.00% on the cross-validation set and 81.20% on an external validation set. The AUC-ROC scores of cross-validations and external validation are 0.95 and 0.85, respectively (Tables 1 and 2).

Clinical toxicity

The clinical toxicity model endpoint reported in this work is defined as whether or not a drug-like molecule failed clinical trials due to toxicity. The datasets were obtained from the ClinTox dataset (30). Data on known clinical toxic and clinical non-toxic drugs/ molecules were defined based on the presence of toxicity reports as withdrawal/failure. kMedoids2—the data sampling method (13) was used to construct the model. The ProTox 3.0 prediction model has a balanced accuracy of 86.00% on the cross-validation set and 79.00% on an external validation set. The AUC-ROC scores of cross-validations and external validation are 0.84 and 0.87, respectively (Tables 1 and 2).

Nutritional toxicity

High-quality toxicological data on food-relevant chemicals were obtained from publicly available databases (31,32) The data was curated on toxic and non-toxic chemicals present in food, and the final dataset was used for training and validation. kMedoids1- the data sampling method (13) was used to construct the model. The ProTox 3.0 predictions model has a balanced accuracy of 83.80% on the cross-validation set and 84.70% on an external validation set. The AUC-ROC scores of cross-validations and external validation are 0.84 and 0.85, respectively (Tables 1 and 2).

表 2. ProTox 3.0 平台中新包含模型在外部验证中的性能评估, 包括准确率、敏感性、特异性和 AUC-ROC

模型 (外部验证)	准确率 (%)	A UC-ROC	敏感性 (%)	特异性 (%)
器官毒性				
神经毒性	80.00	0.91	83.10	78.88
肾毒性	78.00	0.89	83.30	78.90
心脏毒性	89.00	0.95	87.00	88.20
呼吸毒性	88.90	0.93	88.00	85.90
毒理学终点				
血脑屏障 (BBB)	97.10	0.91	96.00	93.00
生态毒性	81.24	0.85	80.00	81.00
临床毒性	79.87	0.87	82.57	78.50
营养毒性	84.70	0.85	73.60	85.60
分子启动事件 (MIE)				
THR α	95.60	0.97	95.60	95.80
THR β	89.50	0.91	70.80	96.20
TTR	85.70	0.94	86.00	85.00
RYR	98.00	0.97	96.00	99.10
加巴尔	94.20	0.98	85.70	94.50
NMDAR	96.50	0.98	89.50	96.90
AMPA受体	98.90	0.97	87.50	99.00
KAR	97.00	0.96	71.90	99.00
AChE	88.50	0.94	89.80	87.60
CAR	99.00	0.98	72.00	98.00
PXR	88.30	0.92	74.10	89.60
纳多克斯	96.00	0.94	81.80	99.30
VGSC	97.10	0.98	87.50	97.60
NIS	99.70	0.99	97.89	98.00
新陈代谢				
CYP1A2	90.00	0.95	84.00	92.00
CYP2C19	95.00	0.87	95.00	95.00
CYP2C9	90.00	0.97	61.00	94.00
CYP2D6	80.00	0.85	80.00	79.00
CYP3A4	86.00	0.93	82.00	87.00
CYP2E1	89.00	0.90	78.00	97.00

毒理学途径

21世纪毒理学 (Tox21) 平台, 美国的毒理学计划, 始于 2008年, 提供了一个包含10000种化学物质数据的库, 这些数据通过高通量实验筛选了针对12种不同基于生物学靶标的通路, 这些通路涉及两大类不良结果通路 (AOP): 核受体通路 (七个基于靶标-通路的模型) 和应激反应通路 (五个基于靶标-通路的模型)。化学化合物在毒理学通路中的活性预测基于Tox21数据集 (33), 这些模型已在我们之前的研究中重新发布 (8)。

分子启动事件 (MIE)

发育期和成人的 衰老神经毒性需要替代方法进行化学品风险评估。由于可能接触到对神经系统具有长期不良健康影响的化合物, 从而导致神经退化性变, 因此制定筛查大量化学品的策略具有高度相关性。不良结局通路 (AOPs) 提供了有关相关分子起始事件 (MIEs) 和关键事件 (KEs) 的信息, 这些信息可能有助于开发针对这些复杂效应的计算替代方法 (34)。

目前在 ProTox 3.0 服务器上开发并托管了 14 个 MIEs 目标端点。关于各自模型、数据点、特征重要性和性能矩阵的更多信息, 可以通过服务器上的模型信息获取。

甲状腺激素受体 α (THR α)、甲状腺激素受体 β (THR β)、转甲状腺素蛋白 (TTR)、

Ryanodine 受体 (RYR)、GABA 受体 (GABAR)、谷氨酸 N-甲基-D-天冬氨酸受体 (NMDAR)、 α -氨基-3-羟基-5-甲基-4-异恶唑丙酸受体 (AMPA)、酪氨酸受体 (KAR)、乙酰胆碱酯酶 (AChE)、本征雄烷受体 (CAR)、孕烷 X 受体 (PXR)、NADH-醌氧化还原酶 (NADHOX)、电压门控钠通道 (VGSC)、Na⁺/I⁻同向转运体 (NIS)。所有模型的平衡准确率均超过 80%, 交叉验证和外部验证的 AUC-ROC 值均在 0.80–0.90 范围内 (见表 1 和表 2)。

新陈代谢

I期代谢可以增加或降低药物的活性, 并且形成的一些代谢物本身可能具有药理活性。本版本的网络服务器包含六种主要的 CYP同工酶, 包括细胞色素CYP1A2、细胞色素CYP2C19、细胞色素CYP2C9、细胞色素CYP2D6、细胞色素CYP3A4、细胞色素CYP2E1, 它们负责临床药物代谢的 >90%。数据集已从不同文献和公开可用的数据库中整理 (35)。有关各自模型、数据点、特征重要性和性能矩阵的更多信息, 可通过服务器上的模型信息获取。

所有模型在交叉验证和外部验证中的平衡准确率均为 >80%, AUC-ROC 值在 0.85–0.98 范围内 (见表 1 和表 2)。

Table 2. Performance evaluation on external validation for the newly included models in the ProTox 3.0 platform in terms of Accuracy, Sensitivity, Specificity and AUC-ROC

Models (external validation)	Accuracy (%)	AUC-ROC	Sensitivity (%)	Specificity (%)
Organ toxicity				
Neurotoxicity	80.00	0.91	83.10	78.88
Nephrotoxicity	78.00	0.89	83.30	78.90
Cardiotoxicity	89.00	0.95	87.00	88.20
Respiratory toxicity	88.90	0.93	88.00	85.90
Toxicological endpoints				
Blood–brain barrier (BBB)	97.10	0.91	96.00	93.00
Ecotoxicity	81.24	0.85	80.00	81.00
Clinical toxicity	79.87	0.87	82.57	78.50
Nutritional toxicity	84.70	0.85	73.60	85.60
Molecular initiating events (MIE)				
THR α	95.60	0.97	95.60	95.80
THR β	89.50	0.91	70.80	96.20
TTR	85.70	0.94	86.00	85.00
RYR	98.00	0.97	96.00	99.10
GABAR	94.20	0.98	85.70	94.50
NMDAR	96.50	0.98	89.50	96.90
AMPAR	98.90	0.97	87.50	99.00
KAR	97.00	0.96	71.90	99.00
AChE	88.50	0.94	89.80	87.60
CAR	99.00	0.98	72.00	98.00
PXR	88.30	0.92	74.10	89.60
NADHOX	96.00	0.94	81.80	99.30
VGSC	97.10	0.98	87.50	97.60
NIS	99.70	0.99	97.89	98.00
Metabolism				
CYP1A2	90.00	0.95	84.00	92.00
CYP2C19	95.00	0.87	95.00	95.00
CYP2C9	90.00	0.97	61.00	94.00
CYP2D6	80.00	0.85	80.00	79.00
CYP3A4	86.00	0.93	82.00	87.00
CYP2E1	89.00	0.90	78.00	97.00

Toxicological pathways

Toxicology in the 21st Century (Tox21) platform, the US toxicology initiative which was started in 2008, provides a library of 10000 chemical data, screened in high-throughput assays against a panel of 12 different biological target-based pathways that involve two major groups of adverse outcome pathways (AOPs): the nuclear receptor pathway (seven target-pathway based models) and the stress response pathway (five target-pathway based models). The prediction of chemical compounds active in toxicological pathways is based on the Tox21 dataset (33), and the models are reposted in our previous studies (8).

Molecular initiating events (MIE)

Developmental and adult/ageing neurotoxicity needs alternative methods for chemical risk assessment. The formulation of a strategy to screen large numbers of chemicals is highly relevant due to potential exposure to compounds that may have long-term adverse health consequences on the nervous system, leading to neurodegeneration. Adverse outcome pathways (AOPs) provide information on relevant molecular initiating events (MIEs) and key events (KEs) that could inform the development of computational alternatives for these complex effects (34).

There are currently 14 MIEs target endpoints developed and hosted on the ProTox 3.0 server. More information on the respective models, data points, feature importance, and performance matrix is available on the server via Model Info.

Thyroid hormone receptor alpha (THR α), Thyroid hormone receptor beta (THR β), Transthyretin (TTR),

Ryanodine receptor (RYR), GABA receptor GABA receptor (GABAR), Glutamate N-methyl-D-aspartate receptor (NMDAR), α -amino-3-hydroxy-5-methyl-4-isoxazolepropionate receptor (AMPA), Kainate receptor (KAR), Acetylcholinesterase (AChE), Constitutive androstane receptor (CAR), Pregnane X receptor (PXR), NADH-quinone oxidoreductase (NADHOX), Voltage-gated sodium channel (VGSC), Na⁺/I⁻ symporter (NIS). All the models have a balanced accuracy of more than 80% and AUC-ROC values within the range of 0.80–0.90 for cross-validation and external validation (see Tables 1 and 2).

Metabolism

Phase I metabolism can increase or decrease the activity of a drug, and some of the metabolites formed may have pharmacological activity themselves. Six major CYP isoforms, including cytochrome CYP1A2, cytochrome CYP2C19, cytochrome CYP2C9, cytochrome CYP2D6, cytochrome CYP3A4, cytochrome CYP2E1, that are responsible for >90% of the metabolism of clinical drugs are included in this version of the web server. Datasets have been curated from different literature and publicly available databases (35). More information on the respective models, data points, feature importance, and performance matrix is available on the server via Model Info.

All the models have a balanced accuracy of >80% and AUC-ROC values within the range of 0.85–0.98 for cross-validation and external validation (see Tables 1 and 2).

应用案例

为了说明 ProTox 3.0 网络服务器的功能及可能的应用，考虑将甲基雄烯酮作为输入化合物。甲基雄烯酮是一种口服活性合成代谢雄激素类固醇。它于1960年投放市场。后来，于1982年从不同国家——法国、德国、英国、美国——撤回；撤回的原因被列为非处方滥用（36）。

使用我们的ProTox 3.0预测流程，甲基睾酮（Methandrostenolone）被预测为急性口服毒性4类，LD50值为1000 mg/kg，预测准确率为100.00%。数据库中报告了三个结构相似的化合物及其理化性质分布图（见图1）。该药物被预测对肝毒性、神经毒性和呼吸毒性具有活性。同时，它还被预测在毒理学终点类别下对血脑屏障通透性、免疫毒性和临床毒性具有活性。四个毒理学通路——NR-AR、NR-AR-LBD、NR-ER和NR-ER-LBD——被预测具有高达1.0的活性概率。一个MIEs终点评价——AcHE——被预测对该药物具有活性（如图1所示）。四个不同的毒性靶点（雄激素受体、胺氧化酶A、糖皮质激素受体和孕酮受体A）被预测具有活性。

结论和未来更新

尽管化学毒性预测领域具有挑战性，但它也可以被视为一个巨大的机遇，无论是在基础科学还是实际应用方面（37）。另一方面，计算方法可以提供更快速且预测精度高的毒性预测。能够将一个化学结构输入软件便获得与体内测试相同的信息，这具有极大的优势。我们已经展示了使用不同采样方法预测新模型毒性在 ProTox 3.0 服务器更新中的进展，涉及 28 个新模型。我们的结果显示，采样方法和基于 RF 的模型在交叉验证和外部验证集上的表现更好（见表 1 和表 2）。在此，我们开发了一个基于机器学习的流程，旨在提高多终点毒性预测的准确性、可解释性和适用性，同时利用现有体外数据的优势。

针对61种不同毒性终点的化合物库。结果可以像在结果页面上报告的那样进行有意义的分析，并且还可以深入了解与某种化合物相关的毒性反应背后的分子机制。这种终点分类也可以帮助药物化学家了解结构与化学毒性之间的关系；这些信息后来可以用于指导先导化合物的结构优化。ProTox分类方法中提出的创新协议可以帮助探索毒性机制。展望未来，收集更多关于AOP的数据也将是必要的，以便通过在生物网络的背景下考虑靶点，从而实现药物作用的机制性理解。网络药理学方法与机器学习方法之间的协同作用可能是未来解决这一问题的潜在途径。

当前版本的 ProTox 在模型中整合了结构警报和毒性基团，但在以有意义的方式可视化这些分子信息方面仍有改进的空间。在未来的更新中，我们致力于增强数据集中结构警报和毒性片段的可视化功能。我们承认可视化表示在促进对数据及其含义的深入理解方面的重要性。通过引入更强大的可视化工具，我们的目标是为用户提供更清晰、更直观的见解，帮助其理解数据集中识别出的结构警报和毒性片段。我们相信，增强可视化方面将大大有助于用户有效地解读和利用 ProTox 提供的信息。

ProTox 网络服务器在普及访问以前仅限于专业软件的复杂预测工具方面发挥着关键作用。通过提供免费可访问的平台，它已经并且还旨在继续为研究人员开放大门，使其能够在药物发现、化学安全、化妆品、食品安全和毒性预测等领域利用先进的计算技术。本质上，像 ProTox 3.0 这样的平台的可用性标志着计算建模包容性迈出的关键一步，使来自不同背景的研究人员能够为科学进步做出贡献。

数据可用性

ProTox 3.0 网络服务器对所有用户免费开放，无需登录，可通过 <https://tox.charite.de> 访问。

补充数据

补充数据可在 NAR 在线获取。

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Application case

To illustrate the functionalities and possible application of the ProTox 3.0 web server, methandrostenolone was considered as the input compound. Methandrostenolone is an orally active anabolic androgenic steroid. It was introduced to the market in 1960. Later, was withdrawn in the year 1982 from different countries—France, Germany, UK, USA; reasons for withdrawal were cited as off-label abuse (36).

Using our ProTox 3.0 prediction pipeline, Methandrostenolone has been predicted with toxicity class 4 for acute oral toxicity with LD₅₀ value of 1000 mg/kg, with a prediction accuracy of 100.00%. Three structurally similar compounds and their physicochemical properties distribution plots are reported from the database (see Figure 1). The drug was predicted to be active for liver toxicity, neurotoxicity, and respiratory toxicity. It was also predicted to be active for the BBB permeability, immunotoxic, and clinically toxic under the toxicological endpoints class. Four toxicological pathways—NR-AR, NR-AR-LBD, NR-ER and NR-ER-LBD—were predicted to be active with a high probability of 1.0. One MIEs endpoint—AcHE was predicted as active for this drug. The metabolic enzyme CYP2C9 was predicted to be active for the drug (as reported in Figure 1). Four different toxicity targets (androgen receptor, amine oxidase A, glucocorticoid receptor and progesterone receptor A) are predicted with probable binding.

Conclusions and future updates

Although the field of prediction of chemical toxicities is challenging, it can also be viewed as one of the great opportunities, both in terms of basic science and practical applications (37). Computational approaches can, on the other hand, provide faster toxicity prediction with great prediction accuracy. The possibility of entering a chemical structure into a piece of software to obtain the same information as from an *in vivo* test is immensely advantageous. We have demonstrated the advance of using different sampling methods to predict the toxicity of 28 new models implemented in the updates of the ProTox 3.0 server. Our results show that sampling methods and RF-based models performed better on cross-validations and external validation sets (See Tables 1 and 2). Here, we have developed a ML-based pipeline with the aim of increasing the accuracy, explainability and applicability of the toxicity predictions for multiple endpoints, taking advantage of the available *in vitro*, *in vivo* data and advanced data sampling methods. Two different molecular representations were tested, Morgan fingerprints and pre-trained maximum common feature (MCF) fingerprints using features derived from MACCS fingerprints. Comparison of the ProTox 3.0 models to that of the published models in literature is provided in the supporting document (Supplementary Table S1). The goal of the ProTox platform is to help users predict chemical toxicity through a computational modelling pipeline and to be considered as one of the alternatives for reducing animal testing and tool for green toxicology applications. The web server is easy to use and favours toxicologists, experimental scientists, and researchers in drug discovery and safer chemical designs. The testimony of the good prediction performance of all the models implemented in the ProTox server is the high performance balanced between sensitivity and specificity. ProTox-web server can help scientific users use the computational screening pipeline to assess the chemical toxicity of large li-

braries of compounds for 61 different toxicity endpoints. The results can be analysed meaningfully as reported on the result page and can further give insights into the molecular mechanism behind such toxic responses associated with a compound. This classification of endpoints can also help medicinal chemists understand the relationship between structure and chemical toxicity; this information can later be used to guide the structure optimization of lead compounds. The innovative protocol presented in the ProTox- classification method can help explore the toxicity mechanism. Moving forward, collecting more data on AOPs will also be essential to allow a mechanistic understanding of drug action by considering targets in the context of biological networks. The synergy between network pharmacology approaches and ML methods could be a potential way of addressing this problem in future.

The current version of ProTox incorporates structural alerts and toxicophores into the models, but there is a need for improvement in visualizing this information in a meaningful manner within the molecules. In our future updates, we are committed to enhancing the visualization aspects of structural alerts and toxic fragments present in our datasets. We acknowledge the significance of visual representations in facilitating a deeper understanding of the data and its implications. By incorporating more robust visualization tools, our aim is to provide users with clearer and more intuitive insights into the structural alerts and toxic fragments identified in our datasets. We believe that enhancing the visualization aspects will greatly benefit users in interpreting and utilizing the information provided by ProTox effectively.

ProTox webserver plays a crucial role in democratizing access to sophisticated prediction tools that were previously confined to specialized software. By offering freely accessible platform, it has been already and also aims to continue open doors for researchers to leverage advanced computational techniques in fields such as drug discovery, chemical safety, cosmetics, food safety, and toxicity prediction. In essence, the availability of platforms like ProTox 3.0 marks a pivotal step towards inclusivity in computational modeling, empowering researchers from diverse backgrounds to contribute to scientific advancements.

Data availability

The ProTox 3.0 web-server is free and open to all users, and there is no login requirement and can be accessed via <https://tox.charite.de>.

Supplementary data

Supplementary Data are available at NAR Online.

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利益冲突声明

未申报。

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Conflict of interest statement

None declared.

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